Coordinate Geometry : Every Question Type

Maths 2 | Std 10 SSC | Chapter 5 | Revision sheet built from your textbook (Practice sets 5.1 to 5.3 and Problem set 5)

Distance	$d(A,B) = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$ From origin: $\sqrt{(x^2 + y^2)}$
Section (m : n)	$\left((m \cdot x_2 + n \cdot x_1)/(m+n) , (m \cdot y_2 + n \cdot y_1)/(m+n) \right)$ *m goes with the 2nd point*
Midpoint	$\left((x_1 + x_2)/2 , (y_1 + y_2)/2 \right)$
Centroid	$\left((x_1 + x_2 + x_3)/3 , (y_1 + y_2 + y_3)/3 \right)$
Slope	$m = (y_2 - y_1)/(x_2 - x_1) = \tan\theta$ | X-axis and lines || X-axis: 0 | Y-axis and lines || Y-axis: not defined | parallel lines: equal slopes
Quick facts	Point on X-axis = (x, 0), on Y-axis = (0, y) | seg || Y-axis: same x | seg || X-axis: same y | Trisection ratios 1:2 and 2:1

TYPE 1 | Distance between two points

Method	Put in the distance formula. Keep brackets around negatives. Leave answer as a surd if not a perfect square.
Example	P(-1,1), Q(5,-7): $d = \sqrt{(6^2 + (-8)^2)} = \sqrt{100} =$ **10**
Practice	Practice set 5.1: Q1 (all six) | Problem set 5: Q6

TYPE 2 | Prove points are collinear

Method	**Slope method (faster):** slope AB = slope BC means collinear. **Distance method:** the two smaller distances add up to the largest.
Example	P(-2,3), Q(1,2), R(4,1): slope PQ = -1/3, slope QR = -1/3, Q is common, so **collinear**. Distance version: A(-3,2), B(1,-2), C(9,-10): $4\sqrt{2} + 8\sqrt{2} = 12\sqrt{2}$.
Practice	Practice set 5.1: Q2 | 5.3: Q3 | Problem set 5: Q2

TYPE 3 | Point on an axis equidistant from two points

Method	On X-axis take (x, 0), on Y-axis take (0, y). Set $PA^2 = PB^2$ (no roots), then solve the linear equation.
Example	Point on Y-axis equidistant from M(-5,-2), N(3,2): $25 + (y+2)^2 = 9 + (y-2)^2$ gives $8y = -16$, $y = -2$. Point is **(0, -2)**.
Practice	Practice set 5.1: Q3 | Problem set 5: Q5

TYPE 4 | Find unknown coordinate from a given distance

Method	Substitute into the distance formula, square both sides, then solve. A squared equation gives **two answers**.
Example	A(2,-2), B(-1,y), AB = 5: $25 = 9 + (y+2)^2$, $(y+2)^2 = 16$, $y+2 = \pm 4$, so **y = 2 or y = -6**.
Practice	Practice set 5.1: Q7

TYPE 5 | Prove a triangle or quadrilateral type

Method	Right triangle: $a^2 + b^2 = c^2$. Equilateral: 3 equal sides. Parallelogram: opposite sides equal. Rectangle: opp. sides equal and diagonals equal. Rhombus: 4 equal sides. Square: 4 equal sides and equal diagonals. Name the points in order and write all lengths.
Example	A(1,7), B(4,2), C(-1,-1), D(-4,4): AB = BC = CD = DA = $\sqrt{34}$ and AC = BD = $\sqrt{68}$, so **square**.
Practice	Practice set 5.1: Q4, Q5, Q6, Q8 | Problem set 5: Q8, Q11, Q12, Q15, Q18

TYPE 6 | Circumcentre / centre and radius of a circle	
Method	Let the point be (a, b). Write $PA^2 = PB^2$ and $PA^2 = PC^2$. Both become linear equations; solve together. Radius = distance from centre to any vertex.
Example	A(-3,-4), B(-5,0), C(3,0): $PA^2 = PB^2$ gives a - 2b = 0; $PA^2 = PC^2$ gives 3a + 2b = -4. Solve: circumcentre **(-1, -1/2)**.
Practice	Problem set 5: Q7, Q16, Q20

TYPE 7 | Find a point dividing a segment (section formula)	
Method	Write which point is (x1, y1) and which is (x2, y2). Use m : n exactly as given. Do x and y separately.
Example	A(3,5), B(7,9), ratio 2:3: x = (2·7 + 3·3)/5 = 23/5, y = (2·9 + 3·5)/5 = 33/5. Point is **(23/5, 33/5)**.
Practice	Practice set 5.2: Q1, Q2 | Problem set 5: Q17

TYPE 8 | Find the ratio in which a point divides a segment	
Method	Let the ratio be m : n, put the known x (or y) in the formula, cross multiply, get m/n. Shortcut: m/n = (x - x1)/(x2 - x). An axis point has x = 0 or y = 0.
Example	A(15,5), B(9,20), P(11,15): 11 = (9m + 15n)/(m+n) gives 2m = 4n, so ratio **2 : 1**.
Practice	Practice set 5.2: Q3, Q5 | Problem set 5: Q4

TYPE 9 | Midpoint and missing endpoint / diameter	
Method	Midpoint = average of coordinates. If the midpoint (or circle centre) is known: 2 × midpoint - known endpoint = missing endpoint.
Example	P(-4,6) divides A(-6,10), B(r,s) in 2:1: -4 = (2r - 6)/3 and 6 = (2s + 10)/3, so **B = (-3, 4)**.
Practice	Practice set 5.2: Q4, Q6 | Problem set 5: Q3

TYPE 10 | Centroid (find centroid, a vertex, or unknowns)	
Method	Centroid = average of the three x and the three y. To find a missing vertex or h, k: put G in the formula and solve.
Example	G(-4,-7), A(-14,-19), B(3,5): x3 = 3(-4) - (-14) - 3 = -1 and y3 = 3(-7) - (-19) - 5 = -7. So **C = (-1, -7)**.
Practice	Practice set 5.2: Q7, Q8, Q9 | Problem set 5: Q14 (midpoints given, centroid is the same)

TYPE 11 | Points of trisection / equal parts	
Method	Trisection: ratios 1:2 and 2:1. Four equal parts: 1:3, 1:1, 3:1. Five parts: 1:4, 2:3, 3:2, 4:1. Or add (B - A)/n repeatedly.
Example	A(2,-2), B(-7,4): P divides 1:2 giving (-1, 0); Q divides 2:1 giving **(-4, 2)**.
Practice	Practice set 5.2: Q10, Q11, Q12 | Problem set 5: Q19

TYPE 12 | Slope of a line or side	
Method	Slope = (y2 - y1)/(x2 - x1). From angle: m = tan θ (45° gives 1, 60° gives √3, 30° gives 1/√3). Same x gives not defined; same y gives 0.
Example	A(-3,5), B(4,-1): m = (-1 - 5)/(4 + 3) = **-6/7**.
Practice	Practice set 5.3: Q1, Q2, Q4 | Problem set 5: Q22

TYPE 13 | Find k using slope, parallel lines, collinearity	
Method	Write the slope with k, equate to the given slope (or to the slope of the parallel line), solve.
Example	P(k,0), Q(-3,-2), slope 2/7: -2/(-3-k) = 2/7 gives **k = 4**.
Practice	Practice set 5.3: Q6, Q7, Q8 | Problem set 5: Q9, Q10

TYPE 14 | Length of medians

Method	Find the midpoint of each opposite side, then use the distance formula from the vertex to that midpoint. Do it three times.
Example	A(-1,1), B(5,-3), C(3,5): midpoint of BC is (4,1), so median from A = $\sqrt{(5^2 + 0^2)}$ = **5**.
Practice	Problem set 5: Q13

TYPE 15 | Fourth vertex of a parallelogram

Method	Diagonals bisect each other, so midpoint of one diagonal = midpoint of the other. Each choice of "opposite pair" gives one D, so there can be **three answers**.
Example	Opposite vertices B and D: D = A + C - B. Repeat with A and C as the other pairs.
Practice	Problem set 5: Q21

TYPE 16 | Fill in the blanks and MCQs (1 mark each)

Method	Facts: seg || Y-axis means same x; point right of origin on X-axis has positive x and y = 0; distance from origin = $\sqrt{(x^2 + y^2)}$; slope = tan θ.
Example	Distance of (-3,4) from origin = $\sqrt{(9 + 16)}$ = **5**.
Practice	Problem set 5: Q1

ANSWER KEY (checked with a calculator script)

Type 1 | Distance between two points
5.1 Q1: (1) 2√2; (2) 4√2; (3) 11/2; (4) 13; (5) 20; (6) 29/2
PS5 Q6: (i) a√2 (ii) 13 (iii) 5a

Type 2 | Prove points are collinear
5.1 Q2: (1) collinear; (2) not collinear; (3) not collinear; (4) collinear
5.3 Q3: (1) collinear; (2) collinear; (3) not collinear; (4) collinear; (5) collinear; (6) collinear
PS5 Q2: (1) collinear; (2) collinear; (3) not collinear

Type 3 | Point on an axis equidistant from two points
5.1 Q3: (-1, 0)
PS5 Q5: (-7, 0)

Type 4 | Find unknown coordinate from a given distance
5.1 Q7: x = 7 or -5

Type 5 | Prove a triangle or quadrilateral type
5.1 Q4: P(-2,2), Q(2,2), R(2,7) : sides 4, 5, √41 : scalene, right-angled ($4^2 + 5^2 = 41$)
5.1 Q5: AB, BC, CD, DA = 5√2, 4√2, 5√2, 4√2 ; diagonals = √82, √82 => rectangle (opp. sides and diagonals equal)
5.1 Q6: AB, BC, CD, DA = 3√10, 3√10, 3√10, 3√10 ; diagonals = 12√2, 6√2 => rhombus (diagonals unequal)
5.1 Q8: sides 4, 4, 4 : equilateral
PS5 Q8: (1) sides √170, √122, 4√10 : scalene | (2) sides 2√5, √5, 3√5 : collinear, NO triangle | (3) sides 4, 4, 4 : equilateral
PS5 Q11: AB, BC, CD, DA = 4√2, √13, 4√2, √13 ; diagonals = √5, √85 => parallelogram (opp. sides equal)
PS5 Q12: AB, BC, CD, DA = √13, 2√13, √13, 2√13 ; diagonals = √65, √65 => rectangle (opp. sides and diagonals equal)
PS5 Q15: AB, BC, CD, DA = √5, √5, √5, √5 ; diagonals = √10, √10 => square
PS5 Q18: AB, BC, CD, DA = √26, √61, √26, √61 ; diagonals = 7, 5√5 => parallelogram (opp. sides equal)

Type 6 | Circumcentre / centre and radius of a circle
PS5 Q7: circumcentre (-1/3, 2/3)
PS5 Q16: circumcentre (25/6, 13/6), radius 13√2/6
PS5 Q20: centre (3, -2)

Type 7 | Find a point dividing a segment (section formula)
5.2 Q1: (1, 3)
5.2 Q2: (1) (-1/3, -1/3) (2) (4/7, -11/7) (3) (0, 13/3)
PS5 Q17: (7, 3)

Type 8 | Find the ratio in which a point divides a segment
5.2 Q3: 2 : 7
5.2 Q5: ratio 2 : 5, k = 6
PS5 Q4: 1 : 3 (Y-axis means x = 0)

Type 9 | Midpoint and missing endpoint / diameter
5.2 Q4: B = (-6, 3)
5.2 Q6: (11, 18)
PS5 Q3: (6, 13)

Type 10 | Centroid (find centroid, a vertex, or unknowns)
5.2 Q7: (1) (1, 3) (2) (6, -2) (3) (19/3, 22/3)
5.2 Q8: C = (-1, -7)
5.2 Q9: h = 7, k = 18
PS5 Q14: (1, 3)

Type 11 | Points of trisection / equal parts
5.2 Q10: (0, 2), (-2, -3)
5.2 Q11: (-9, -8), (-4, -6), (1, -4)
5.2 Q12: (16, 12), (12, 14), (8, 16), (4, 18)
PS5 Q19: A (20, 10), P (16, 12), R (8, 16), B (0, 20)

Type 12 | Slope of a line or side
5.3 Q1: (1) 1 (2) √3 (3) not defined
5.3 Q2: (1) 2; (2) -3/8; (3) 5/2; (4) 5/4; (5) 1/2; (6) not defined
5.3 Q4: AB = -5, BC = 1/5, CA = -2/3
PS5 Q22: AC = 10, BD = 0

Type 13 | Find k using slope, parallel lines, collinearity
5.3 Q6: k = 5
5.3 Q7: k = 0
5.3 Q8: k = 5
PS5 Q9: k = 5
PS5 Q10: slope AB = -3, slope CD = -3, so AB || CD

Type 14 | Length of medians
PS5 Q13: medians = 5, 2√13, √37

Type 15 | Fourth vertex of a parallelogram
PS5 Q21: D = (7, 6), (3, 6) or (-1, -10)

Type 16 | Fill in the blanks and MCQs (1 mark each)
PS5 Q1: (1) D (1, -3) (2) D (2, 0) (3) C (5) (4) C (1/√3)

Note: Practice set 5.1 Q5 points actually form a rectangle (diagonals also equal), which is still a parallelogram. Show only what the question asks.